

EE4415 Lab Assignment
Lab1_Project_1 – Synthesizing a 16-bit RISC CPU

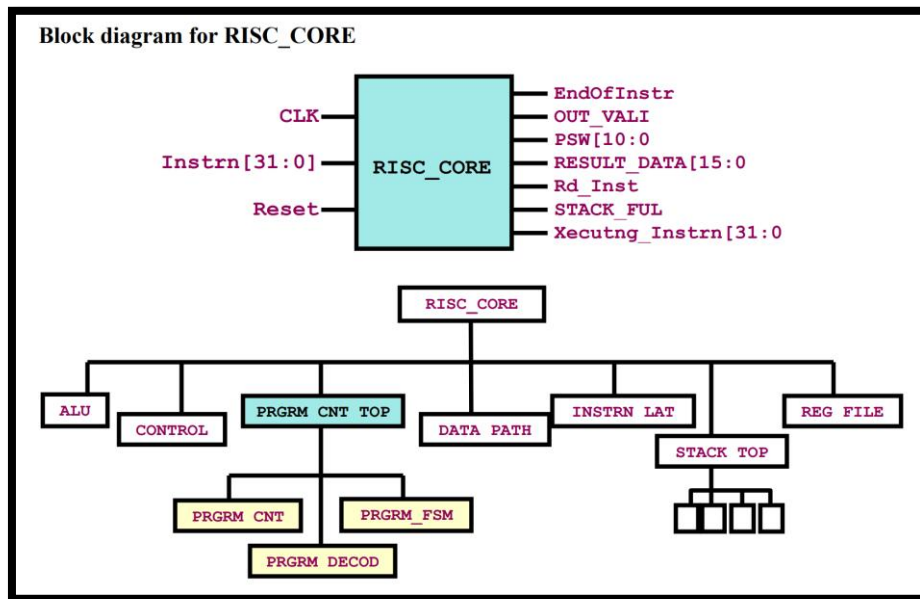


Fig 1: RISC_Core Block Diagram

1. Unit 1: Verilog, VCS & DVE

Objective: To learn to design the PRGRM_CNT sub-module of the RISC_Core in Verilog, simulate the code using VCS Simulator and viewing waveform in DVE.

1.1 Design PRGRM_CNT in Verilog

cd ~/ee4415_part1

gedit prgrm_cnt.v

```
module PRGRM_CNT (Reset, Clk, Incrmnt_PC, Ld_Brnch_Addr, Ld_Rtn_Addr, Imm_Addr, Return_Addr, PC);
input Reset, Clk, Incrmnt_PC, Ld_Brnch_Addr, Ld_Rtn_Addr;
input [7:0] Imm_Addr, Return_Addr;
output [7:0] PC; //wire
reg [7:0] PC_reg; // storage
assign PC = PC_reg;

//Program Counter update (priority: Reset > Increment > Return > Branch)
always @(posedge Clk or posedge Reset) begin
    if (Reset) begin
        PC_reg <= 8'd0; //PC=0 when Reset=1
    end
    else if (Incrmnt_PC) begin
        PC_reg <= PC_reg + 8'd1; //PC value increments by 1 when Incrmnt_PC=1
    end
    else if (Ld_Rtn_Addr) begin
        PC_reg <= Return_Addr; //Return_Addr value's store into PC when Ld_Rtn_Addr=1
    end
    else if (Ld_Brnch_Addr) begin
        PC_reg <= Imm_Addr; //Imm_Addr value's store into PC when Ld_Brnch_Addr=1
    end
    else begin
        PC_reg <= PC; // hold value (optional; can be modified)
    end
end
endmodule
```

Fig 2: Code snippet of PRGRM_CNT

PRGRM_CNT Verilog file completed with specifications mentioned below:

- PC set to 0 when Reset is 1
- PC is incremented by 1 when Incrmnt_PC is 1
- PC is set to Return_Addr when Ld_Rtn_Addr is 1
- PC is set to Imm_Addr when Ld_Brnch_Addr is 1

1.2 Compile and simulate prgrm_cnt.v

Run following command to compile prgrm_cnt.v and prgrm_cnt_tb.v to generate simv file.

vcs -full64 +lint=all -debug_all -timescale=1ns/10ps prgrm_cnt.v prgrm_cnt.tb.v

Check .simv file generated using “ls” command

Run simv file using the command: `./simv`

```
[a329221m@lxa30 ~/ee4415_part1]$ vcs -full64+lint=all -debug_all -timescale=1ns/10ps prgrm_cnt.v prgrm_cnt_tb.v
```

```
Warning:[DEBUG_DEP] Option will be deprecated
The option '-debug_all' will be deprecated in a future release. Please use
'-debug_acc+all -debug_region+cell+encrypt' instead.

Chronicologic VCS (TM)
Version Q-2020.03-SP2-Full64 -- Mon Mar 2 00:37:36 2026
Copyright (c) 1991-2020 by Synopsys Inc.
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This program is proprietary and confidential information of Synopsys Inc.
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controlling such use and disclosure.

Warning:[DEBUG_DEP] Option will be deprecated
The option '-debug_all' will be deprecated in a future release. Please use
'-debug_acc+all -debug_region+cell+encrypt' instead.

Parsing design file 'prgrm_cnt.v'
Parsing design file 'prgrm_cnt_tb.v'
Top Level Modules:
    PRGRM_CNT_TB
TimeScale is 1 ns / 10 ps
Starting vcs inline pass...

1 module and 0 UDP read.

However, due to incremental compilation, no re-compilation is necessary.
/app1/synopsys/vcs/Q-2020.03-SP2/bin/vcs: line 32726: 3865051 Segmentation fault
(core dumped) $(TOOL_HOME)/bin/cfs_ident_exec -f $(XML_INPU
T_EXE) -o $(SEARCHDIR)/idents_tapt.xml -o SrcFile "${dir$Srcfiles}/src_files.c" ${all_dyn_libs} > tapt.xml_writer.log
rm -f cuarc*.so csrc*.so pre_vcsobj*.so share_vcsobj*.so
ld -shared -Bsymbolic -no-relax -o .././simv.daidir/?.cuarc0.so objs/AmcQw_d.o
rm -f cuarc0.so
if [ x -x ./simv ]; then chmod a-x ./simv; fi
g++ -o ./simv -rdynamic -Wl,-rpath=$ORIGIN/?.simv.daidir -Wl,-rpath=../simv.daidir -Wl,-rpath=/app1/synopsys/vcs/Q-2020.03-SP2/linux64/lib
-L/app1/synopsys/vcs/Q-2020.03-SP2/linux64/lib -Wl,-rpath-link=../usr/lib64/libnuma.so.1 -l3864902_archive.1.so -lprev_archive.1.so -lcuarc0
.so -lSIML0 -lrmmapats.mop.o -lrmmapats.o -lrmr.o -lrmr_nd.o -lrmr_llvm_0.1.o -lrmr_llvm_0_0.o -lrvrsim -lerronrl -ltnspnmalloc -ltnspn -lv
csnew -ltnspnrefile -ltnclutative -lapp1/synopsys/vcs/Q-2020.03-SP2/linux64/lib/vcs_tls.o -Wl,-whole-archive -lvcuactrl -Wl,-no-ldvs -lvs
/app1/synopsys/vcs/Q-2020.03-SP2/linux64/lib/vcs_save_restore_new.o -ldl -lc -lm -lpthread -ldl
./simv up to date
CPU time: .185 seconds to compile + .230 seconds to elab + .172 seconds to link
```

```

[a329221m@xa30 ~]/ee4415_part1$ ls
alias          default.svf    PRGRM_CNT.ddc  results.vpd    Synopsys_stack_trace_3044648.txt  tcl_procs
command.log    DVEfiles      PRGRM_CNT_DW01_inc_0.ddc  run.log        syn_timing_1.rpt                    tmpfile1212
constraint.tcl  ee4415_2526_lab1_part1  PRGRM_CNT.rpt  run.tcl        syn_timing_2.rpt                    ucli.key
crte_003044648.txt  Previews     prgrm_cnt_tb.v  simv           syn_timing_3.rpt                    vrt
csrc           prgrm_cnt.ddc  prgrm_cnt.v    simv.daidir    syn_timing.rpt

[a329221m@xa30 ~]/ee4415_part1$ ./simv
Chronologic VCS simulator copyright 1991-2020
Contains Synopsys proprietary information.
Compiler version 0-2020.03-SP2_Full64; Runtime version 0-2020.03-SP2_Full64; Mar  2 00:38 2026
VCD+ Writer 0-2020.03-SP2_Full64 Copyright (c) 1991-2020 by Synopsys Inc.

Warning-[VPD-WARN-VCD-OFF] Warning from $vcdplusoff.
At time 2900, in PLI routine called from prgrm_cnt_tb.v, 33
  Stopping the recording of signal value changes for specified scopes or
  specific signals may cause unpredictable values on other dumped signals.
  Turn off dumping on the whole design to guarantee that the value changes
  being dumped are correct.

$finish called from file "prgrm_cnt_tb.v", line 35.
$finish at simulation time          2900
          VCS Simulation Report

Time: 20000 ps
CPU Time: 0.250 seconds;      Data structure size:  0.0Mb
Mon Mar  2 00:38:41 2026
[a329221m@xa30 ~]/ee4415_part1$ dve -full64 -vpd results.vpd &
[1] 3865204

```

Fig 3: Compilation and Simulation snippets

1.3 Waveform

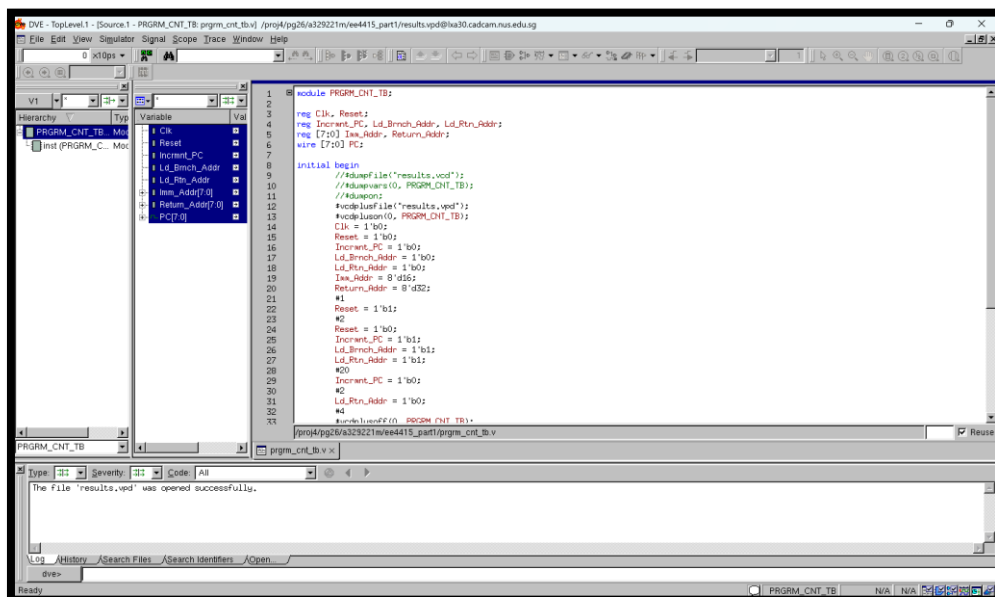
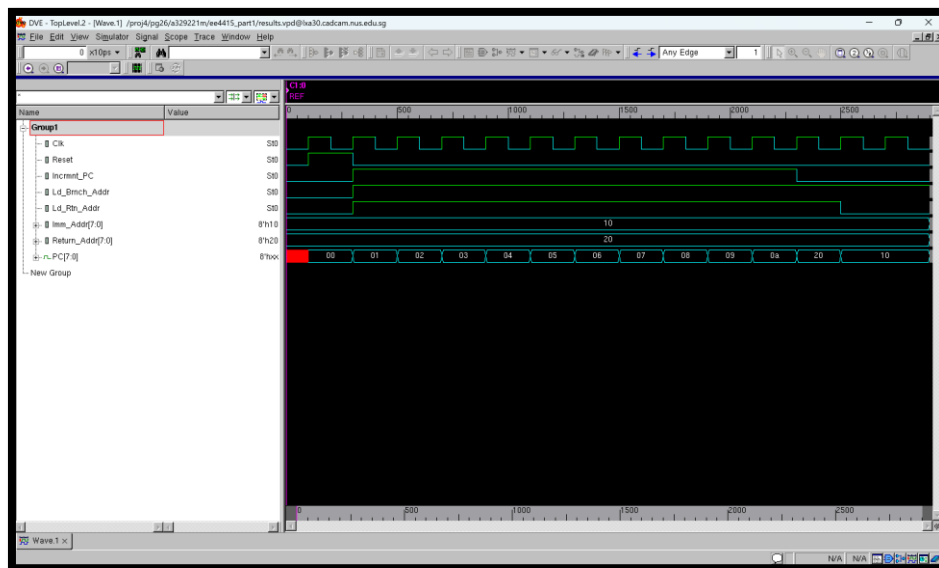
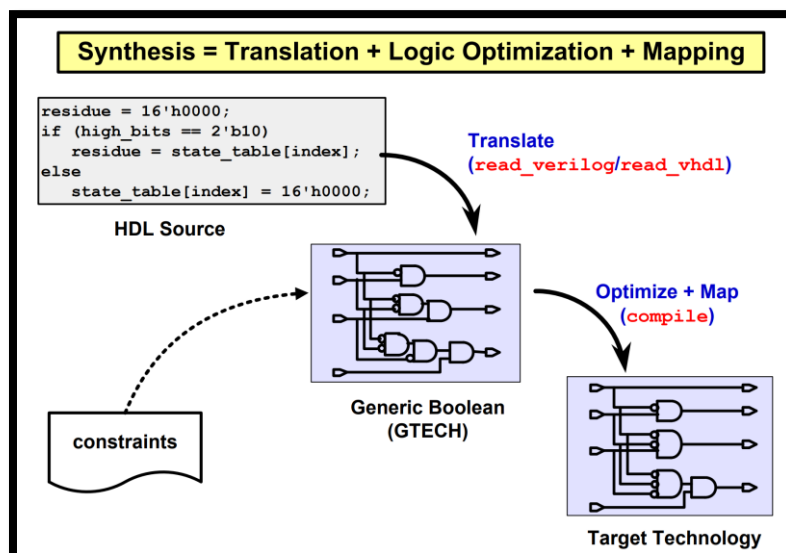


Fig 4: results.vpd file



2. Unit 2: Introduction to Synopsys Design Flow

Objective: The unit describes basic steps involved in the synthesis process and how to navigate through the design's hierarchy using Design Vision, the graphical interface to Design Compiler.



2.1 Invoke Design Vision

First come to working directory:

cd ~/ee4415 part1

It is the same directory in which .synopsys dc.setup file is present.

Use below command to start Design Vision:

/app11/synopsys/syn/P-2019.03-SP4/bin/design vision-xg

2.2 Read RTL

```
./app11/synopsys/syn/P-2019.03-SP4/libraries/dw/sim ver
```

```
[a329221m@lxa30 ~/ee4415_part1]$ /app11/synopsys/syn/P-2019.03-SP4/bin/design_vision-xg

Design Compiler Graphical
DC Ultra (TM)
DFTMAX (TM)
Power Compiler (TM)
DesignWare (R)
DC Expert (TM)
Design Vision (TM)
HDL Compiler (TM)
VHDL Compiler (TM)
DFT Compiler
Design Compiler(R)

Version P-2019.03-SP4 for linux64 - Aug 28, 2019

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of a written license agreement with Synopsys, Inc. All other use, reproduction,
or distribution of this software is strictly prohibited.

Initializing...
Initializing gui preferences from file /proj4/pg26/a329221m/.synopsys_dv_prefs.tcl
design_vision> 4.1
design_vision> read_file -format verilog {/proj4/pg26/a329221m/ee4415_part1/prgrm_cnt.v}
Loading db file '/app11/saed_32.28/SAED32_EDK/lib/stdcell_rvt/db_ccs/saed32rvt_tt1p05v125c.db'
Information: Using CCS timing libraries. (TIM-024)
Loading db file '/app11/synopsys/syn/P-2019.03-SP4/libraries/syn/gtech.db'
Loading db file '/app11/synopsys/syn/P-2019.03-SP4/libraries/syn/standard.sldb'
Loading link library 'saed32rvt_tt1p05v125c'
Loading link library 'gtech'
Loading verilog file '/proj4/pg26/a329221m/ee4415_part1/prgrm_cnt.v'
Detecting input file type automatically (-rtl or -netlist).
Reading with Presto HDL Compiler (equivalent to -rtl option).
Running PRESTO HDLC
Compiling source file /proj4/pg26/a329221m/ee4415_part1/prgrm_cnt.v
```

```
Inferred memory devices in process
in routine PRGRM_CNT line 10 in file
'/proj4/pg26/a329221m/ee4415_part1/prgrm_cnt.v'.

=====
| Register Name | Type | Width | Bus | MB | AR | AS | SR | SS | ST |
=====
| PC_reg_reg | Flip-flop | 8 | Y | N | Y | N | N | N | N |
=====

Presto compilation completed successfully.
Current design is now '/proj4/pg26/a329221m/ee4415_part1/PRGRM_CNT.db:PRGRM_CNT'
Loaded 1 design.
Current design is 'PRGRM_CNT'.
design_vision> Current design is 'PRGRM_CNT'.
```

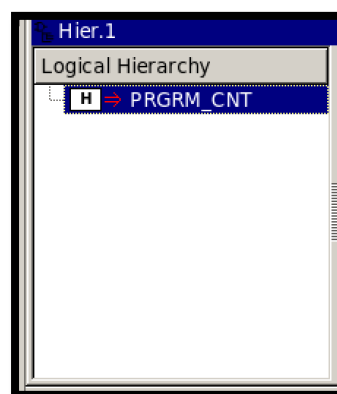


Fig 7: Verilog file opened in Design Vision

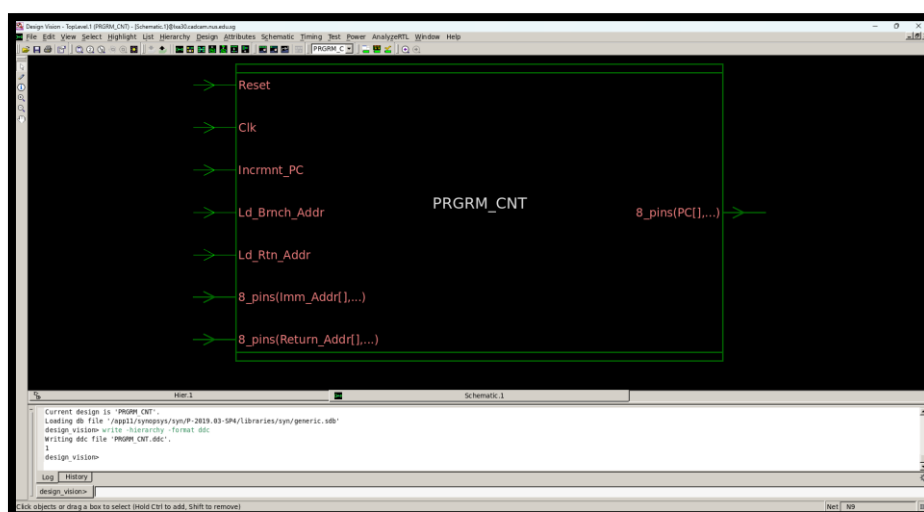


Fig 8: High-level schematic view

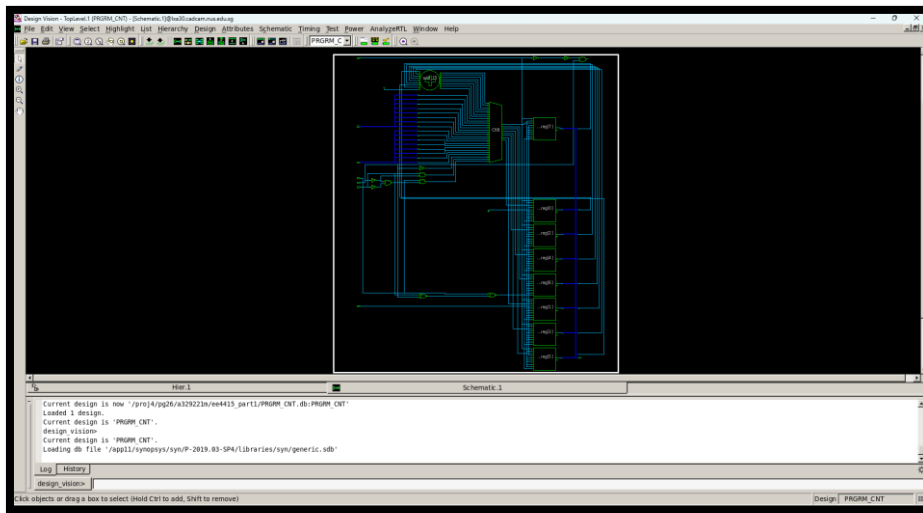


Fig 9: PRGRM_CNT Block

2.3 Apply Constraints:

Complete constraint.tcl file referring to the specifications mentioned below in Fig (ix) and apply the constraints using “source constraint.tcl” command.

Clock period	1.5 ns
Clock uncertainty	0.1 ns
Max input delay (except Clk port)	1 ns
Max output delay	1 ns
Driving cell and pin at the input	DFFX1_RVT, pin Q
Output load	10 times the capacitance seen at the input port A of the INVX0_RVT cell
Max area	200

Fig 10: Constraint Specifications

```
#####
# constraint.tcl (EE4415 Lab Part 1 - Task 3)
#####

# Clock (port name in your Verilog is: Clk)
set CLK_PORT CLK
set CLK_NAME CLK
set PERIOD 1.5
set CLK_UNC 0.1
set IN_DLY 1.0
set OUT_DLY 1.0
set MAX_AREA 200

# 1) Create clock + uncertainty
create_clock -name $CLK_NAME -period $PERIOD [get_ports $CLK_PORT]
set_clock_uncertainty -setup $CLK_UNC [get_clocks $CLK_NAME]

# 2) Input constraints (exclude clock port)
set IN_PORTS [remove_from_collection [all_inputs] [get_ports $CLK_PORT]]
set_input_delay -max $IN_DLY -clock $CLK_NAME $IN_PORTS
set_driving_cell -lib_cell DFFX1_RVT -pin Q $IN_PORTS

# 3) Output constraints
set_output_delay -max $OUT_DLY -clock $CLK_NAME [all_outputs]

# Output load = 10x cap seen at INVX0_RVT/A
set INV_A_PIN [get_lib_pins */INVX0_RVT/A]
set OUT_LOAD [expr 10.0 * [load_of $INV_A_PIN]]
set_load $OUT_LOAD [all_outputs]

# 4) Max area
set_max_area $MAX_AREA
```

Fig 11: constraint.tcl code snippet

For checking if constraints are set correctly use below commands:

```
design_vision> report_units
*****
Report : units
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Mon Mar  2 01:09:35 2026

*****
Units
-----
Time_unit           : 1.0e-09 Second(ns)
Capacitive_load_unit : 1.0e-15 Farad(fF)
Resistance_unit      : 1.0e+6 Ohm(MOhm)
Voltage_unit         : 1 Volt
Power_unit           : N/A
Current_unit         : 1.0e-06 Amp(uA)
1
```

Fig 12: view report_lib

```
design_vision> report_clock-skew-attributes
Information: Changed wire load model for 'DW01_inc_width8' from '(none)' to 'ForQA'. (OPT-170)
Information: Changed wire load model for 'ADD_UN5_OP_8_1_8' from '(none)' to 'ForQA'. (OPT-170)
Information: Updating design information... (UID-85)
*****
Report : clocks
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Mon Mar  2 01:12:22 2026
*****
Attributes:
d - dont_touch_network
f - fix_hold
p - propagated_clock
g - generated_clock
g - lib_generated_clock

Clock      Period  Waveform      Attrs      Sources
-----
CLK        1.50   {0 0.75}      {clk}

*****
Report : clock skew
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Mon Mar  2 01:12:22 2026
*****
Object      Rise      Fall      Min Rise  Min fall  Uncertainty
            Delay    Delay    Delay     Delay     Plus      Minus
-----
CLK         -      -      -         -         -        0.10
1
Current design is 'PRGRM_CNT'.
```

Fig 13: report_clock-skew-attributes

```
design_vision> report_port-verbose
*****
Report : port
-verbose
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Mon Mar  2 01:13:25 2026
*****
Port      Dir      Pin Load  Wire Load  Max Trans  Max Cap  Connection Class  Attrs
-----
Clk       in        0.0000  0.0000    --         --       --
Imm_Addr[0] in        0.0000  0.0000    --         208.00  --
Imm_Addr[1] in        0.0000  0.0000    --         208.00  --
Imm_Addr[2] in        0.0000  0.0000    --         208.00  --
Imm_Addr[3] in        0.0000  0.0000    --         208.00  --
Imm_Addr[4] in        0.0000  0.0000    --         208.00  --
Imm_Addr[5] in        0.0000  0.0000    --         208.00  --
Imm_Addr[6] in        0.0000  0.0000    --         208.00  --
Imm_Addr[7] in        0.0000  0.0000    --         208.00  --
Incrmnt_PC in        0.0000  0.0000    --         208.00  --
Ld_Brnch_Addr in      0.0000  0.0000    --         208.00  --
Ld_Rtn_Addr in      0.0000  0.0000    --         208.00  --
Reset     in        0.0000  0.0000    --         208.00  --
Return_Addr[0] in      0.0000  0.0000    --         208.00  --
Return_Addr[1] in      0.0000  0.0000    --         208.00  --
Return_Addr[2] in      0.0000  0.0000    --         208.00  --
Return_Addr[3] in      0.0000  0.0000    --         208.00  --
Return_Addr[4] in      0.0000  0.0000    --         208.00  --
Return_Addr[5] in      0.0000  0.0000    --         208.00  --
Return_Addr[6] in      0.0000  0.0000    --         208.00  --
Return_Addr[7] in      0.0000  0.0000    --         208.00  --
PC[0]     out       5.5372  0.0000    --         --       --
PC[1]     out       5.5372  0.0000    --         --       --
PC[2]     out       5.5372  0.0000    --         --       --
PC[3]     out       5.5372  0.0000    --         --       --
PC[4]     out       5.5372  0.0000    --         --       --
PC[5]     out       5.5372  0.0000    --         --       --
PC[6]     out       5.5372  0.0000    --         --       --
PC[7]     out       5.5372  0.0000    --         --       --
```

Port	External Number Points	Max Wireload Model	Min Wireload Model	Min Pin Load	Min Wire Load
Clk	1	--	--	--	--
Imm_Addr[0]	1	--	--	--	--
Imm_Addr[1]	1	--	--	--	--
Imm_Addr[2]	1	--	--	--	--
Imm_Addr[3]	1	--	--	--	--
Imm_Addr[4]	1	--	--	--	--
Imm_Addr[5]	1	--	--	--	--
Imm_Addr[6]	1	--	--	--	--
Imm_Addr[7]	1	--	--	--	--
Incrmnt_PC	1	--	--	--	--
Ld_Brnch_Addr	1	--	--	--	--
Ld_Rtn_Addr	1	--	--	--	--
Reset	1	--	--	--	--
Return_Addr[0]	1	--	--	--	--
Return_Addr[1]	1	--	--	--	--
Return_Addr[2]	1	--	--	--	--
Return_Addr[3]	1	--	--	--	--
Return_Addr[4]	1	--	--	--	--
Return_Addr[5]	1	--	--	--	--
Return_Addr[6]	1	--	--	--	--
Return_Addr[7]	1	--	--	--	--
PC[0]	1	--	--	--	--
PC[1]	1	--	--	--	--
PC[2]	1	--	--	--	--
PC[3]	1	--	--	--	--
PC[4]	1	--	--	--	--
PC[5]	1	--	--	--	--
PC[6]	1	--	--	--	--
PC[7]	1	--	--	--	--

Input Port	Input Delay		Max	Related Clock	Max Fanout
	Rise	Fall			
Clk	--	--	--	--	--
Imm_Addr[0]	--	--	1.00	CLK	--
Imm_Addr[1]	--	--	1.00	CLK	--
Imm_Addr[2]	--	--	1.00	CLK	--
Imm_Addr[3]	--	--	1.00	CLK	--
Imm_Addr[4]	--	--	1.00	CLK	--
Imm_Addr[5]	--	--	1.00	CLK	--
Imm_Addr[6]	--	--	1.00	CLK	--
Imm_Addr[7]	--	--	1.00	CLK	--
Incrmnt_PC	--	--	1.00	CLK	--
Ld_Brnch_Addr	--	--	1.00	CLK	--
Ld_Rtn_Addr	--	--	1.00	CLK	--
Reset	--	--	1.00	CLK	--
Return_Addr[0]	--	--	1.00	CLK	--
Return_Addr[1]	--	--	1.00	CLK	--
Return_Addr[2]	--	--	1.00	CLK	--
Return_Addr[3]	--	--	1.00	CLK	--
Return_Addr[4]	--	--	1.00	CLK	--
Return_Addr[5]	--	--	1.00	CLK	--
Return_Addr[6]	--	--	1.00	CLK	--
Return_Addr[7]	--	--	1.00	CLK	--

Input Port	Driving Cell			Attrs(min/max)
	Rise(min/max)	Fall(min/max)	Mult(min/max)	
Imm_Addr[0]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[1]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[2]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[3]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[4]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[5]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[6]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Imm_Addr[7]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Incrmnt_PC	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Ld_Brnch_Addr	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Ld_Rtn_Addr	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Reset	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[0]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[1]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[2]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[3]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[4]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[5]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[6]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	
Return_Addr[7]	DFFX1_RVT/Q	DFFX1_RVT/Q	-- / --	

Input Port	Max Drive		Min Drive		Resistance		Min Cap	Min Fanout	Cell Deg
	Rise	Fall	Rise	Fall	Max	Min			
Clk	--	--	--	--	--	--	--	--	--
Imm_Addr[0]	--	--	--	--	--	--	--	--	--
Imm_Addr[1]	--	--	--	--	--	--	--	--	--
Imm_Addr[2]	--	--	--	--	--	--	--	--	--
Imm_Addr[3]	--	--	--	--	--	--	--	--	--
Imm_Addr[4]	--	--	--	--	--	--	--	--	--
Imm_Addr[5]	--	--	--	--	--	--	--	--	--
Imm_Addr[6]	--	--	--	--	--	--	--	--	--
Imm_Addr[7]	--	--	--	--	--	--	--	--	--
Incrmnt_PC	--	--	--	--	--	--	--	--	--
Ld_Brnch_Addr	--	--	--	--	--	--	--	--	--
Ld_Rtn_Addr	--	--	--	--	--	--	--	--	--
Reset	--	--	--	--	--	--	--	--	--
Return_Addr[0]	--	--	--	--	--	--	--	--	--
Return_Addr[1]	--	--	--	--	--	--	--	--	--
Return_Addr[2]	--	--	--	--	--	--	--	--	--
Return_Addr[3]	--	--	--	--	--	--	--	--	--
Return_Addr[4]	--	--	--	--	--	--	--	--	--
Return_Addr[5]	--	--	--	--	--	--	--	--	--
Return_Addr[6]	--	--	--	--	--	--	--	--	--
Return_Addr[7]	--	--	--	--	--	--	--	--	--

Input Port	Max Tran		Min Tran	
	Rise	Fall	Rise	Fall
Clk	--	--	--	--
Imm_Addr[0]	--	--	--	--
Imm_Addr[1]	--	--	--	--
Imm_Addr[2]	--	--	--	--
Imm_Addr[3]	--	--	--	--
Imm_Addr[4]	--	--	--	--
Imm_Addr[5]	--	--	--	--
Imm_Addr[6]	--	--	--	--
Imm_Addr[7]	--	--	--	--
Incrmnt_PC	--	--	--	--
Ld_Brnch_Addr	--	--	--	--
Ld_Rtn_Addr	--	--	--	--
Reset	--	--	--	--
Return_Addr[0]	--	--	--	--
Return_Addr[1]	--	--	--	--
Return_Addr[2]	--	--	--	--
Return_Addr[3]	--	--	--	--
Return_Addr[4]	--	--	--	--
Return_Addr[5]	--	--	--	--
Return_Addr[6]	--	--	--	--
Return_Addr[7]	--	--	--	--

```
design_vision> report_port -verbose [get_ports PC*]
```

Report : port
 Design : PRGRM_CNT
 Version: P-2019.03-SP4
 Date : Mon Mar 2 01:18:08 2026

Port	Dir	Pin Load	Wire Load	Max Trans	Max Cap	Connection Class	Attrs
PC[7]	out	5.5372	0.0000	--	--	--	--
PC[6]	out	5.5372	0.0000	--	--	--	--
PC[5]	out	5.5372	0.0000	--	--	--	--
PC[4]	out	5.5372	0.0000	--	--	--	--
PC[3]	out	5.5372	0.0000	--	--	--	--
PC[2]	out	5.5372	0.0000	--	--	--	--
PC[1]	out	5.5372	0.0000	--	--	--	--
PC[0]	out	5.5372	0.0000	--	--	--	--

Port	External Number Points	Max Wireload Model	Min Wireload Model	Min Pin Load	Min Wire Load
PC[7]	1	--	--	--	--
PC[6]	1	--	--	--	--
PC[5]	1	--	--	--	--
PC[4]	1	--	--	--	--
PC[3]	1	--	--	--	--
PC[2]	1	--	--	--	--
PC[1]	1	--	--	--	--
PC[0]	1	--	--	--	--

Output Port	Output Delay		Related		Fanout Load
	Min Rise	Max Fall	Min Rise	Max Fall	
PC[7]	--	--	1.00	1.00	CLK 0.00
PC[6]	--	--	1.00	1.00	CLK 0.00
PC[5]	--	--	1.00	1.00	CLK 0.00
PC[4]	--	--	1.00	1.00	CLK 0.00
PC[3]	--	--	1.00	1.00	CLK 0.00
PC[2]	--	--	1.00	1.00	CLK 0.00
PC[1]	--	--	1.00	1.00	CLK 0.00
PC[0]	--	--	1.00	1.00	CLK 0.00

1
 design_vision> get_attribute [get_ports PC[0]] load
 5.537200

Figs 14-21 report_port -verbose

2.4 Compile

Compile the design using the “compile” command.

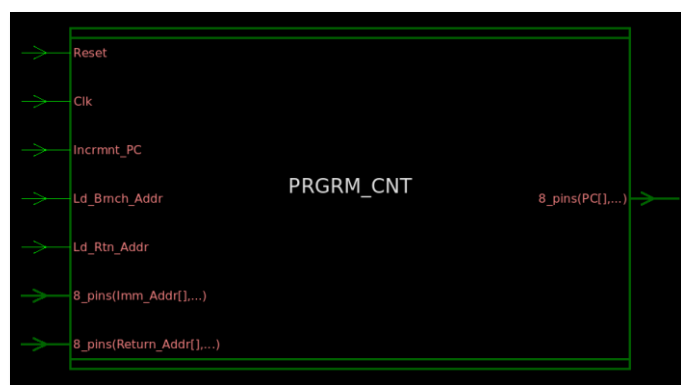


Fig 22: High-level compiled schematic

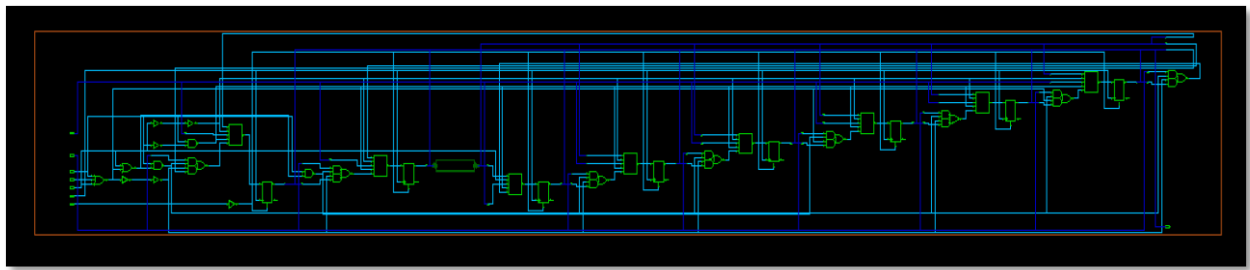


Fig 23: PRGRM_CNT compiled schematic

2.5 Analyze Timing

Create Path Slack Histogram

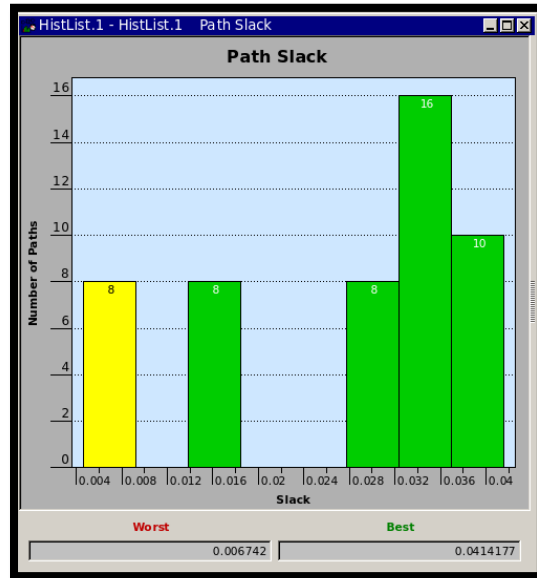


Fig 24: Path Slack Histogram

2.6 Generate Timing and Area Report

Command: report_timing

```
design_vision> report_timing

*****
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 12:31:38 2026
*****

Operating Conditions: tt1p05v125c  Library: saed32rvt_tt1p05v125c
Wire Load Model Mode: enclosed

Startpoint: Ld_Rtn_Addr
             (input port clocked by CLK)
Endpoint:   PC_reg_reg[0]
             (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type:  max
```

Des/Clust/Port	Wire Load Model	Library	
PRGRM_CNT	ForQA	saed32rvt_tt1p05v125c	
Point	Incr	Path	
clock CLK (rise edge)	0.00	0.00	
clock network delay (ideal)	0.00	0.00	
input external delay	1.00	1.00 r	
Ld_Rtn_Addr (in)	0.01	1.01 r	
U32/Y (NOR3X2_RVT)	0.09	1.10 f	
U23/Y (NOR2X0_RVT)	0.07	1.17 r	
U20/Y (AND2X1_RVT)	0.06	1.23 r	
U19/Y (AO22X1_RVT)	0.06	1.28 r	
U18/Y (AO221X1_RVT)	0.06	1.34 r	
PC_reg_reg[0]/D (DFFARX1_RVT)	0.01	1.36 r	
data arrival time		1.36	
clock CLK (rise edge)	1.50	1.50	
clock network delay (ideal)	0.00	1.50	
clock uncertainty	-0.10	1.40	
PC_reg_reg[0]/CLK (DFFARX1_RVT)	0.00	1.40 r	
library setup time	-0.04	1.36	
data required time		1.36	
data arrival time		-1.36	
slack (MET)		0.01	

Figs 25-26 Timing Report

Timing Report Summary:

- Max delay (data arrival time / path delay): 1.36
- Slack: 0.01 (MET)

the increments are:

- clock network delay (ideal): 0.00 → Path 0.00
- input external delay: +1.00 → Path 1.00

- iii. Ld_Rnt_Addr (in): +0.01 → Path 1.01
- iv. U32/Y: +0.09 → Path 1.10
- v. U23/Y: +0.07 → Path 1.17
- vi. U20/Y: +0.06 → Path 1.23
- vii. U19/Y: +0.06 → Path 1.29
- viii. U18/Y: +0.06 → Path 1.35
- ix. PC_reg_reg[0]/D: +0.01 → Path 1.36

So the calculation is:

$$\text{Data arrival time} = 1.00 + 0.01 + 0.09 + 0.07 + 0.06 + 0.06 + 0.06 + 0.01 = 1.36 \text{ ns}$$

Command: report_area

```
design_vision> report_area
*****
Report : area
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 12:35:12 2026
*****

Library(s) Used:

    saed32rvt_tt1p05v125c (File: /app11/saed_32.28/SAED32_EDK/lib/stdcell_rvt/db_ccs/saed32rvt_tt1p05v125c.db)

Number of ports:          45
Number of nets:           86
Number of cells:          44
Number of combinational cells: 35
Number of sequential cells:  8
Number of macros/black boxes: 0
Number of buf/inv:        7
Number of references:      9

Combinational area:       89.458688
Buf/Inv area:             8.895848
Noncombinational area:    56.928257
Macro/Black Box area:     0.000000
Net Interconnect area:    13.348816

Total cell area:          146.386945
Total area:               159.735761
1
```

Fig 27: Area Report

Area Report Summary:

- **Total area = 159.735761** (library units)
- **Total cell area = 146.386945**

Area and slack values are given below:

- **Area:** 159.735761
- **Slack:** 0.01 (MET)

Important Info:

Specifying an Area Goal

```
dc_shell-xg-t> current_design PRGRM_CNT_TOP
dc_shell-xg-t> set_max_area 100
```

- Units are defined by the target library:
 - 2-input-NAND-gate
 - Transistors
 - Square microns

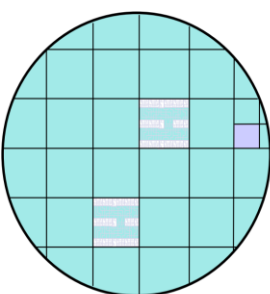


Fig 28: Area units

```
design_vision> report_units
*****
Report : units
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 12:42:06 2026
*****

Units
-----
Time_unit           : 1.0e-09 Second(ns)
Capacitive_load_unit : 1.0e-15 Farad(fF)
Resistance_unit      : 1.0e+6 Ohm(MOhm)
Voltage_unit         : 1 Volt
Power_unit           : N/A
Current_unit         : 1.0e-06 Amp(uA)
1
```

Fig 29: Report Units

Slack	Startpoint Pin Name	Endpoint Pin Name	Path Group	Endpoint Clock Skew	Arrival	Required	Startpoint Clock Name	Endpoint Clock Name
0.00674	Ld_Rtn_Addr	PC_reg_reg[0]/D	CLK	0.00000	1.35606	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[1]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[2]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[3]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[4]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[5]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00681	Ld_Rtn_Addr	PC_reg_reg[6]/D	CLK	0.00000	1.35599	1.36280	CLK	CLK
0.00683	Ld_Rtn_Addr	PC_reg_reg[7]/D	CLK	0.00000	1.35599	1.36282	CLK	CLK
0.14606	PC_reg_reg[0]/CLK	PC[0]	CLK	0.00000	0.25394	0.40000	CLK	CLK
0.15585	PC_reg_reg[7]/CLK	PC[7]	CLK	0.00000	0.24415	0.40000	CLK	CLK
0.15591	PC_reg_reg[6]/CLK	PC[6]	CLK	0.00000	0.24409	0.40000	CLK	CLK
0.15591	PC_reg_reg[5]/CLK	PC[5]	CLK	0.00000	0.24409	0.40000	CLK	CLK
0.15591	PC_reg_reg[4]/CLK	PC[4]	CLK	0.00000	0.24409	0.40000	CLK	CLK
0.15591	PC_reg_reg[3]/CLK	PC[3]	CLK	0.00000	0.24409	0.40000	CLK	CLK
0.15591	PC_reg_reg[2]/CLK	PC[2]	CLK	0.00000	0.24409	0.40000	CLK	CLK
0.15591	PC_reg_reg[1]/CLK	PC[1]	CLK	0.00000	0.24409	0.40000	CLK	CLK

Fig 30: Timing Analysis Driver

The Critical path is highlighted in Fig (31):

Starting point Pin Name: **Ld_Rtn_Addr**

End point Pin Name: **PC_reg_reg[0]/D**

Critical Path is the one with the minimum slack value which is **0.00674** as highlighted in Fig (30).

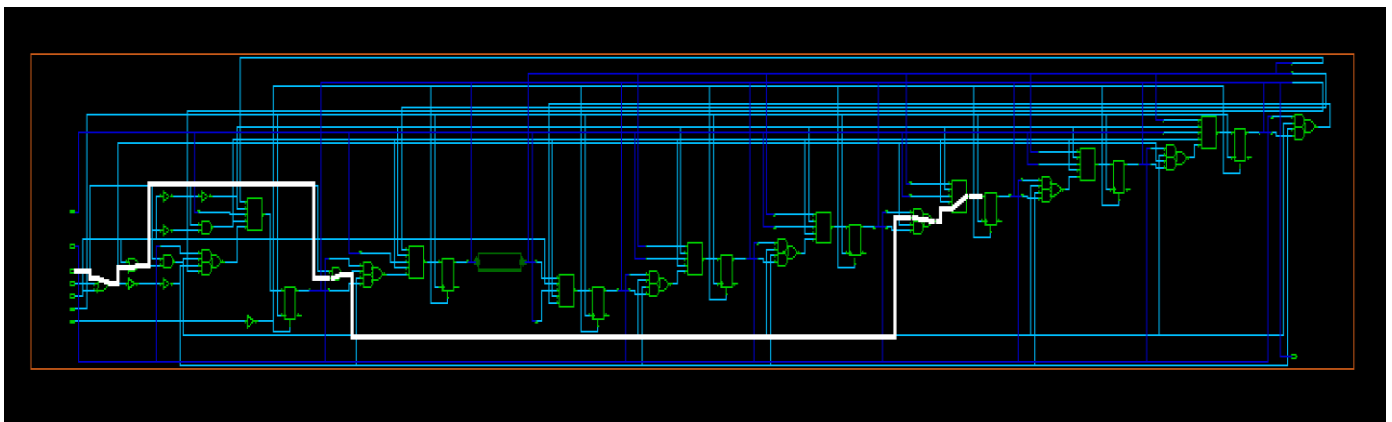


Fig 31: Critical Path

2.7 Save Mapped Design

File => Save As

Filename: prgrm_cnt.ddc

2.8 Remove Designs from Design Compiler Memory

File => Remove All Designs

2.9 Exit Design Vision

File => Exit => OK

3. Unit 3: Introduction to DC-Tcl

3.1 Automating part of synthesis flow by writing a Tcl script and running script using Design Compiler.

3.2 Steps involved:

- Read file prgrm_cnt.v
- Set current design to PRGRM_CNT
- Perform a link
- Constrain the design by applying the constraint.tcl file
- Compile the design
- Generate and save the results from a constraint report to PRGRM_CNT.rpt using following command: **redirect -tee PRGRM_CNT.rpt {report_constraint -all_violators}**
- Save the mapped design as PRGRM_CNT.ddc
- Quit Design Compiler

```
Beginning Delay Optimization Phase
-----

```

ELAPSED TIME	AREA	WORST NEG SLACK	TOTAL SETUP COST	DESIGN RULE COST	ENDPOINT
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	

```
Beginning Area-Recovery Phase (max_area 200)
-----

```

ELAPSED TIME	AREA	WORST NEG SLACK	TOTAL SETUP COST	DESIGN RULE COST	ENDPOINT
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	
0:00:01	140.5	0.00	0.0	0.0	

Note: Symbol # after min delay cost means estimated hold TNS across all active scenarios

```

Optimization Complete
-----
1
# 6) Report constraints into PRGRM_CNT.rpt
redirect -tee PRGRM_CNT.rpt {report_constraint -all_violators;
Information: Updating design information... (UID-85)

*****
Report : constraint
        -all_violators
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 15:28:33 2026
*****

1
# 7) Save mapped design as PRGRM_CNT.ddc
write -format ddc -hierarchy -output PRGRM_CNT.ddc
Writing ddc file 'PRGRM_CNT.ddc'.

1
# 8) Quit
quit

Thank you...
[a329221m@lxa30 ~/ee4415_part1]$ grep Error run.log

```

Figs 32-37 DC Tcl code snippets

3.3 Post completing run.tcl file, execute it using following command:

```
/app11/synopsys/syn/P-2019.03-SP4/bin/dc shell-xg-t -f run.tcl | tee -> run.log
```

```

#complete this file according to the requirements in Unit 3

# 1) Reading prgrm_cnt.v file
read_verilog prgrm_cnt.v

# 2) Set current design
current_design PRGRM_CNT

# 3) Link
link

# 4) Apply Constraints
source ./constraint.tcl

# 5) Compile
compile

# 6) Report constraints into PRGRM_CNT.rpt
redirect -tee PRGRM_CNT.rpt {report_constraint -all_violators}

# 7) Save mapped design as PRGRM_CNT.ddc
write -format ddc -hierarchy -output PRGRM_CNT.ddc

# 8) Quit
quit

```

Fig 38: Run.tcl code snippet

3.4 Check run.log visually to verify there are no errors in the file.

Command: `grep Error run.log`

3.5 Verify files PRGRM_CNT.ddc and PRGRM_CNT.rpt are successfully created.

```

Information: Updating design information... (UID-85)

*****
Report : constraint
        -all_violators
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 15:28:33 2026
*****

1

```

Fig 39: 2 files generated: PRGRM_CNT.ddc & PRGRM_CNT.rpt

4. Unit 4: Timing Reports

4.1 Load Mapped PRGRM_CNT Design

Command: `/app11/synopsys/syn/P-2019.03-SP4/bin/dc_shell-xg-t`

```

[a329221m@lxa30 ~/ee4415_part1]$ /app11/synopsys/syn/P-2019.03-SP4/bin/dc_shell-xg-t

Design Compiler Graphical
DC Ultra (TM)
DFTMAX (TM)
Power Compiler (TM)
DesignWare (R)
DC Expert (TM)
Design Vision (TM)
HDL Compiler (TM)
VHDL Compiler (TM)
DFT Compiler
Design Compiler(R)

Version P-2019.03-SP4 for linux64 - Aug 28, 2019

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Inc. This software may only be used in accordance with the terms and conditions
of a written license agreement with Synopsys, Inc. All other use, reproduction,
or distribution of this software is strictly prohibited.
Initializing...
Initializing gui preferences from file /proj4/pg26/a329221m/.synopsys_dv_prefs.tcl

```

Fig 40: Loaded mapped design

4.2 Generate and Interpret 4 timing reports

(I) **Question 1.** Are there any unconstrained timing paths in PRGRM_CNT?

Use Command: `check_timing` to verify unconstrained timing paths.

```

Current design is 'PRGRM_CNT'.
design_vision> check_timing
Information: Updating design information... (UID-85)
Information: Checking generated_clocks...
Information: Checking loops...
Information: Checking no_input_delay...
Information: Checking unconstrained_endpoints...
Information: Checking pulse_clock_cell_type...
Information: Checking no_driving_cell...
Information: Checking partial_input_delay...
Warning: there are 20 input ports that only have partial input delay specified. (TIM-212)
-----
Reset
Incrmnt_PC
Ld_Brnch_Addr
Ld_Rtn_Addr
Imm_Addr[7]
Imm_Addr[6]
Imm_Addr[5]
Imm_Addr[4]
Imm_Addr[3]
Imm_Addr[2]
Imm_Addr[1]
Imm_Addr[0]
Return_Addr[7]
Return_Addr[6]
Return_Addr[5]
Return_Addr[4]
Return_Addr[3]
Return_Addr[2]
Return_Addr[1]
Return_Addr[0]
1

```

Fig 41: check_timing report

There are no unconstrained timing paths in PRGRM_CNT. There is just a warning for 20 ports as listed in the snap above having partial input delay.

How many path groups are in PRGRM_CNT?

Use the command: report_path_group

```

report_path_group

*****
Report : path_group
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 16:15:50 2026
*****

Group Name      Weight      Critical
-----
**default**     1.00      0.00
CLK             1.00      0.00

Path Group CLK:
    -to      CLK

```

Fig 42: path_group_report

There are 2 path groups: default and CLK

(II) Use view utility to generate default timing report and answer the following questions.

Commands: read_ddc PRGRM_CNT.ddc

report_timing > syn_timing.rp

Reading ddc file '/proj4/pg26/a329221m/ee4415_part1/PRGRM_CNT_DW01_inc_0.ddc'.			
Information: Updating design information... (UID-85)			

Report : timing			
-path full			
-delay max			
-max_paths 1			
Design : PRGRM_CNT			
Version: P-2019.03-SP4			
Date : Thu Feb 19 16:47:10 2026			

Operating Conditions: tt1p05v125c		Library: saed32rvt_tt1p05v125c	
Wire Load Model Mode: enclosed			
Startpoint: Ld_Rtn_Addr			
(input port clocked by CLK)			
Endpoint: PC_reg_reg[0]			
(rising edge-triggered flip-flop clocked by CLK)			
Path Group: CLK			
Path Type: max			

Des/Clust/Port	Wire Load Model	Library	
PRGRM_CNT	ForQA	saed32rvt_tt1p05v125c	

Point	Incr	Path	

clock CLK (rise edge)	0.00	0.00	
clock network delay (ideal)	0.00	0.00	
input external delay	1.00	1.00 r	
Ld_Rtn_Addr (in)	0.01	1.01 r	
U32/Y (NOR3X2_RVT)	0.09	1.10 f	
U23/Y (NOR2X0_RVT)	0.07	1.17 r	
U20/Y (AND2X1_RVT)	0.06	1.23 r	
U19/Y (AO22X1_RVT)	0.06	1.28 r	
U18/Y (AO221X1_RVT)	0.06	1.34 r	
PC_reg_reg[0]/D (DFFARX1_RVT)	0.01	1.36 r	
data arrival time		1.36	

clock CLK (rise edge)	1.50	1.50	
clock network delay (ideal)	0.00	1.50	
clock uncertainty	-0.10	1.40	

PC_reg_reg[0]/CLK (DFFARX1_RVT)	0.00	1.40 r
library setup time	-0.04	1.36
data required time		1.36

data required time		1.36
data arrival time		-1.36

slack (MET)		0.01

Fig 43: Syn_timing report

Question 3: Is this a setup time or hold time timing report?

Setup timing report, as it is a **max delay** report (-delay max, Path Type: max). Hold checks use **min delay**.

Question 4. What is the start point (input port or clk pin of internal register)?

Startpoint is **Ld_Rtn_Addr**, and the report clearly says “**input port clocked by CLK**”

Question 5. What is the end point?

PC_reg_reg[0] (a rising-edge triggered flip-flop), specifically its **D capture point**:

PC_reg_reg[0]/D (DFFARX1_RVT).

Question 6. Under what operating conditions was this timing report generated?

- **Operating Conditions:** tt1p05v125c
- **Library:** saed32rvt_tt1p05v125c

Wire Load Model Mode: enclosed

WLM used = ForQA

The timing report is being calculated for **typical process, 1.05 V supply, at 125°C**

Question 7. Did the timing path meet or violate it's constraint?

The timing path meets it's constraint as slack shows **0.01** → positive slack means it passes.

Question 8. What is the clock period for Clk?

The clock period for Clk is **1.50 ns**. It is shown by the capture clock edge line: clock CLK (rise edge) 1.50

Question 9. What is the “input external delay” and where did this number come from?

Input external delay = 1.00 ns is the **input delay constraint** applied at the input port relative to the clock (from constraints.tcl file).

It comes from: set_input_delay 1.0 -clock CLK <input_ports>

DC adds this before the internal logic delay when computing **data arrival time**.

Question 10. Does the design's partitioning break a combinational path. If so explain?

The timing path is **input** → **combinational logic** → **register (flip-flop)**.

The **flip-flop PC_reg_reg[0]** forms a **timing boundary**, so the combinational logic is “broken” (terminated) at that register for setup analysis.

Question 11. What is the setup time requirement of the capture register?

From the report: library setup time -0.04

So the **setup time requirement magnitude is 0.04 ns (40 ps)**.

(Shown negative because it is **subtracted** from the available clock period in the required-time calculation.)

Question 12. What does the clock uncertainty number represent?

Clock uncertainty (**0.10 ns** here, shown as -0.10) is a **timing margin** that accounts for:

- clock jitter
- clock skew
- other clock variation/uncertainty

It is subtracted from the clock period to make timing more conservative.

(III) Use view utility to generate timing report with input pins and with 6 significant digits.

Commands: read_ddc PRGRM_CNT.ddc

report_timing input_pins -significant 6 > syn_timing_1.rpt

```
Reading ddc file '/proj4/pg26/a329221m/ee4415_part1/PRGRM_CNT_DW01_inc_0.ddc'.
Information: Updating design information... (UID-85)

*****
Report : timing
        -path full
        -delay max
        -input_pins
        -max_paths 1
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 18:21:01 2026
*****

Operating Conditions: tt1p05v125c  Library: saed32rvttt1p05v125c
Wire Load Model Mode: enclosed

Startpoint: Ld_Rtn_Addr
            (input port clocked by CLK)
Endpoint:  PC_reg_reg[0]
            (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: max
```

Des/Clust/Port	Wire Load Model	Library	
PRGRM_CNT	ForQA	saed32rvttt1p05v125c	
Point	Incr	Path	
clock CLK (rise edge)	0.000000	0.000000	
clock network delay (ideal)	0.000000	0.000000	
input external delay	1.000000	1.000000 r	
Ld_Rtn_Addr (in)	0.014006	1.014006 r	
U32/A1 (NOR3X2_RVT)	0.016662	1.030668 r	
U32/Y (NOR3X2_RVT)	0.072085	1.102753 f	
U23/A2 (NOR2X0_RVT)	0.019887	1.122640 f	
U23/Y (NOR2X0_RVT)	0.047212	1.169853 r	
U20/A2 (AND2X1_RVT)	0.015866	1.185719 r	
U20/Y (AND2X1_RVT)	0.040439	1.226158 r	
U19/A4 (AO22X1_RVT)	0.017253	1.243411 r	
U19/Y (AO22X1_RVT)	0.039642	1.283053 r	
U18/A5 (AO221X1_RVT)	0.012938	1.295991 r	
U18/Y (AO221X1_RVT)	0.047311	1.343302 r	
PC_reg_reg[0]/D (DFFARX1_RVT)	0.012759	1.356061 r	
data arrival time		1.356061	
clock CLK (rise edge)	1.500000	1.500000	
clock network delay (ideal)	0.000000	1.500000	
clock uncertainty	-0.100000	1.400000	
PC_reg_reg[0]/CLK (DFFARX1_RVT)	0.000000	1.400000 r	
library setup time	-0.037196	1.362803	
data required time		1.362803	
data required time		1.362803	
data arrival time		-1.356061	
slack (MET)		0.006742	

Figs 44-45: Syn_timing_1 report

Question 13. What is different in this timing report from the default one?

This report includes **input pins** in the timing path (shows pin-level points like U32/A1, U32/A2, etc.) and it prints timing numbers with **6 significant digits**, so the delays/slack are much more precise than the default report.

(IV) Use the view utility to generate timing report with net names and fanout.

Commands: read_ddc PRGRM_CNT.ddc

report_timing vrt -nets > syn_timing_2.rpt

Report : timing			
-path full			
-delay max			
-nets			
-max_paths 1			
Design : PRGRM_CNT			
Version: P-2019.03-SP4			
Date : Thu Feb 19 18:21:43 2026			

Operating Conditions: tt1p05v125c Library: saed32rvrt_tt1p05v125c			
Wire Load Model Mode: enclosed			
Startpoint: Ld_Rtn_Addr			
(input port clocked by CLK)			
Endpoint: PC_reg_reg[0]			
(rising edge-triggered flip-flop clocked by CLK)			
Path Group: CLK			
Path Type: max			
Des/Clust/Port Wire Load Model Library			

PRGRM_CNT	ForQA	saed32rvrt_tt1p05v125c	
Point	Fanout	Incr	Path

clock CLK (rise edge)		0.00	0.00
clock network delay (ideal)		0.00	0.00
input external delay		1.00	1.00 r
Ld_Rtn_Addr (in)		0.01	1.01 r
Ld_Rtn_Addr (net)	4	0.00	1.01 r
U32/Y (NOR3X2_RVT)		0.09	1.10 f
n3 (net)	4	0.00	1.10 f
U23/Y (NOR2X0_RVT)		0.07	1.17 r
n12 (net)	3	0.00	1.17 r
U20/Y (AND2X1_RVT)		0.06	1.23 r
n4 (net)	4	0.00	1.23 r
U19/Y (AO22X1_RVT)		0.06	1.28 r
n11 (net)	1	0.00	1.28 r
U18/Y (AO221X1_RVT)		0.06	1.34 r
n22 (net)	1	0.00	1.34 r
PC_reg_reg[0]/D (DFFARX1_RVT)		0.01	1.36 r
data arrival time			1.36

clock CLK (rise edge)		1.50	1.50
clock network delay (ideal)		0.00	1.50
clock uncertainty		-0.10	1.40
PC_reg_reg[0]/CLK (DFFARX1_RVT)		0.00	1.40 r
library setup time		-0.04	1.36
data required time			1.36

data required time			1.36
data arrival time			-1.36

slack (MET)			0.01

Figs 46-47: Syn_timing_2 report

Question 14. What delay is associated with each net and why is the delay 0?

In -nets timing report, the net lines are the ones that look like Ld_Rtn_Addr_net, n3, n12, n4, n11, n22, etc. For each net line, the incremental delay (Incr) shown is 0.00.

It is zero as Synopsys DC's report_timing breaks the delay into *cell (arc) delay* at the pin/cell lines. The net name line is just a connectivity label (showing the net and fanout), so DC reports 0 incremental delay on the net line itself. Also, in this synthesis timing report (pre-layout), there is no extracted interconnect RC, so net delay is not modeled as a separate incremental timing item here. So,

- Net delay shown on net-name rows = 0.00
- Zero because net rows are informational; timing increment is accounted on the cell/pin arcs, and pre-layout has no real routed net RC.

Question 15. What does fanout column represent?

Fanout represents the number of loads (sinks) driven by a particular net.

Meaning: how many input pins/instances a particular net connects to.

For eg:

- Ld_Rtn_Addr_net has Fanout = 4 → that net drives 4 loads.
- n12 has Fanout = 3 → that net drives 3 loads.
- n11 has Fanout = 1 → that net drives 1 load.

(V) Use the view utility to generate a timing report for hold.

Commands: read_ddc PRGRM_CNT.ddc

report_timing vrt -delay min > syn_timing_3.rpt

```
*****
Report : timing
        -path full
        -delay min
        -max_paths 1
Design : PRGRM_CNT
Version: P-2019.03-SP4
Date   : Thu Feb 19 18:22:11 2026
*****

Operating Conditions: tt1p05v125c  Library: saed32rvt_tt1p05v125c
Wire Load Model Mode: enclosed

Startpoint: PC_reg_reg[0]
              (rising edge-triggered flip-flop clocked by CLK)
Endpoint: PC_reg_reg[0]
              (rising edge-triggered flip-flop clocked by CLK)
Path Group: CLK
Path Type: min

Des/Clust/Port    Wire Load Model    Library
-----
PRGRM_CNT         ForQA              saed32rvt_tt1p05v125c

Point                                     Incr      Path
-----
clock CLK (rise edge)                    0.00      0.00
clock network delay (ideal)              0.00      0.00
PC_reg_reg[0]/CLK (DFFARX1_RVT)          0.00      0.00 r
PC_reg_reg[0]/Q (DFFARX1_RVT)           0.13      0.13 r
add_15/A[0] (PRGRM_CNT_DW01_inc_0)       0.00      0.13 r
add_15/U1/Y (INVX1_RVT)                  0.03      0.16 f
add_15/SUM[0] (PRGRM_CNT_DW01_inc_0)     0.00      0.16 f
U18/Y (AO221X1_RVT)                      0.07      0.22 f
PC_reg_reg[0]/D (DFFARX1_RVT)           0.01      0.24 f
data arrival time                        0.24
-----
clock CLK (rise edge)                    0.00      0.00
clock network delay (ideal)              0.00      0.00
PC_reg_reg[0]/CLK (DFFARX1_RVT)          0.00      0.00 r
library hold time                       -0.01     -0.01
data required time                      -0.01
-----
data required time                      -0.01
data arrival time                      -0.24
-----
slack (MET)                             0.24
*****
```

Fig 48: Syn_timing_3 report

Question 16. Is this a setup or hold time report?

It is Hold time report as it is generated with -delay min and shows Path Type: min (min delay is used for hold checks).

Question 17. What is the start point?

Startpoint: PC_reg_reg[0].

It is a rising edge-triggered flip-flop clocked by CLK (launch register).

Question 18. What is the end point?

Endpoint: PC_reg_reg[0]

Also a rising edge-triggered flip-flop clocked by CLK (capture register). So this is a reg-to-reg (same register) hold path in the report.

Question 19. Did this timing path meet or violate its constraints?

It met the constraint. Slack is **0.24** → positive slack means it passes.

Question 20. Under what operating conditions was this report generated?

- **Operating Conditions:** tt1p05v125c
- **Library:** saed32rvt_tt1p05v125c
- **Wire Load Model Mode** = enclosed
- **WLM** used = ForQA

The timing report is being calculated for **typical process, 1.05 V supply, at 125°C**

Question 21. Is this an appropriate operating condition for hold time calculations?

No, usually hold time is checked at the **fast corner** (minimum delays), like **FF / fast / high V / low T**. The report uses **TT (typical-typical)**, which is not the most pessimistic for hold.

Question 22. What is the hold time requirement of the end point?

From the report line: library hold time **-0.01**

The hold time requirement magnitude is **0.01 ns (10 ps)**.

In other words: **Hold requirement = 0.01 ns** (reported as **-0.01** because it's applied in the required-time calculation).

Question 23. What is the delay through the launching register?

From the timing report, the launching FF is **PC_reg_reg[0]** and the line:

PC_reg_reg[0]/Q (DFDFFAX1_RVT) Incr = 0.13 ns

is the **clock-to-Q delay** of the launching register.

So, Delay through launching register = 0.13 ns

Question 24. Is this delay enough to satisfy hold time?

This report is a **hold check** (-delay min).

- library hold time = -0.01
- slack (MET) = +0.24

Since hold slack is **positive**, the hold requirement is satisfied.